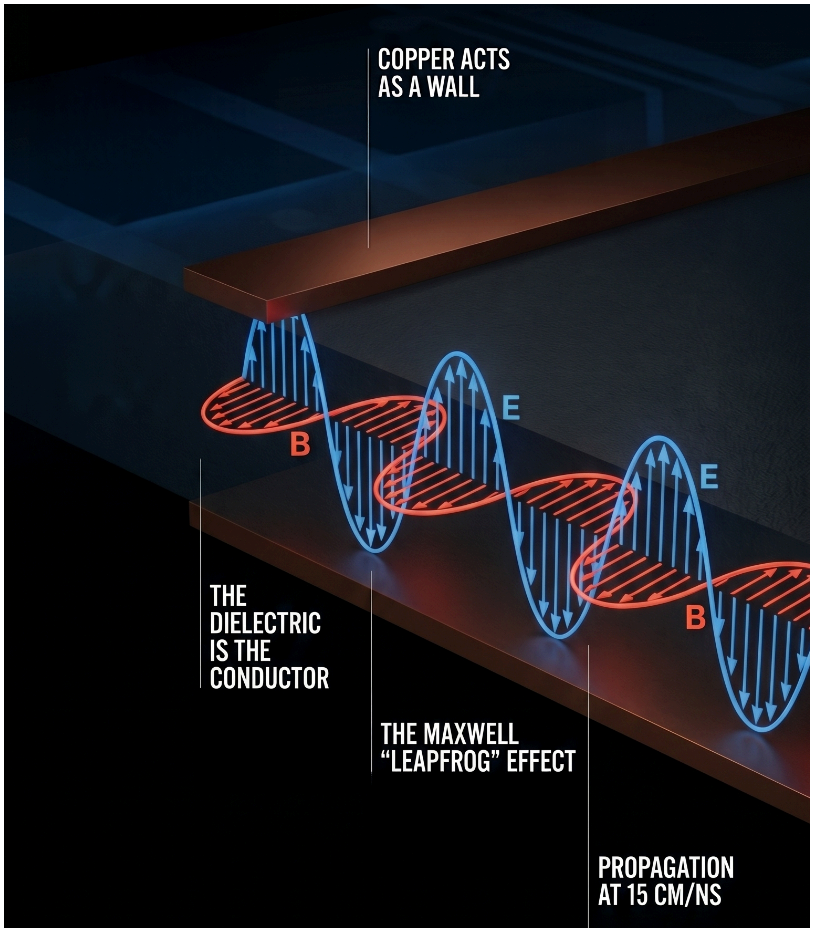
Field Theory for PCB Design

"Signal Integrity from First Principles"



This document develops the electromagnetic theory behind PCB trace design and derives practical layout rules from it. It is a companion to the Board Design Guide, providing the physical reasoning that underpins the design decisions made there.

Chapter 1 builds the field-theory picture from first principles — starting with Maxwell's equations applied to a microstrip, and working through wave propagation, conductor response, rail collapse, crosstalk, and EMI. Chapter 2 translates that physics into concrete layout rules, stack-up choices, and trace sizing for OPNhydro.

The treatment assumes familiarity with basic circuit theory (Ohm's law, Kirchhoff's laws) but does not require a background in electromagnetics. Where the mathematics goes deeper, expandable sections and Feynman-style narratives offer the same insight from a different angle.

Most textbook introductions to transmission lines start with the lumped-element model — an infinite ladder of series inductors and shunt capacitors. That model yields the right equations (the wave equation, $Z_0 = \sqrt{L/C}$, propagation delay), but it obscures the physics. It cannot explain why a split in the ground plane causes a signal integrity problem, why energy travels in the dielectric rather than in the copper, or why crosstalk depends on field geometry rather than circuit topology. This document takes a different path: it starts from the fields themselves — Maxwell's equations applied to the cross-section of a microstrip — and derives the circuit quantities (impedance, capacitance, inductance) as consequences. The result is a mental model that transfers directly to layout decisions, where the fields, not the lumped elements, are what the designer actually controls.

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1. Field Theory

Circuit Theory, the simplified low-frequency approximation taught in undergraduate courses, is based on field theory. It assumes that the physical size of components is much smaller than the wavelength of the signal, so we can ignore wave propagation and use simple circuit laws — Ohm's Law, Kirchhoff's laws. Before the mid-1990s, a typical device might output signals with 10 ns rise times at 10 MHz — and the circuits worked with the crudest of interconnects.

This chapter builds up the field-theory picture from first principles:

- **§1.1** — how a voltage step becomes an EM wave, and where its energy actually flows.
- **§1.2** — what the copper does in response: sustained currents and surface-charge confinement.
- **§1.3** — how the conduction current in the copper and the displacement current in the dielectric sustain one continuous magnetic field.
- **§1.4** — what happens when the power distribution network cannot supply current fast enough (rail noise).
- **§1.5** — how the fields of one trace leak into another (crosstalk).
- **§1.6** — what happens when the field escapes the board entirely (EMI).

1.1. EM Wave in the Dielectric

> "Energy and signals travel in the spaces not the traces" -- Ralph Morrison

As clock frequencies increase, rise times shorten — as a rule of thumb, the rise time t_r is roughly 10% of the clock period:

$$t_r \approx \frac{1}{10\, f_{clk}}$$

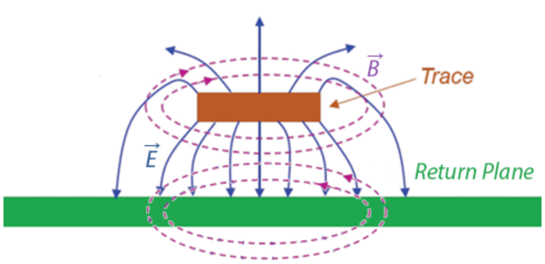
A 100 MHz clock demands 1 ns. At these short rise times, the wavelength of the signal's frequency content becomes comparable to (within ~10×) the physical dimensions of the PCB traces, and the circuit-theory assumptions fail. **Field Theory** is now required to account for radiation, retardation and wave propagation.

PCB design shifts from drawing paths for current to designing transmission lines that contain EM fields, manage parasitic inductance and capacitance, and control radiated emissions. Those physics classes about electromagnetic fields and transmission lines are no longer reserved for RF engineers — they become practical for anyone designing a fast PCB.

A microstrip has **two regions**: the dielectric between the conductors, and the copper itself. §1.1 looks at the dielectric — that is where the wave lives and where the energy flows. §1.2 then turns to the copper, where the free electrons respond.

From Voltage to EM Wave

> "Electromagnetic (EM) field theory, based on Maxwell's equations, is the fundamental description of electrical phenomena (fields, waves, radiation)." -- Dr. Eric Bogatin [1] and Kenneth Wyatt [2].



Cross-section view of Microstrip fields.
(Courtesy: Patrick André)

> **A note on field lines.** Textbook diagrams show fields as lines with arrows. These *field lines* are a visualization invented by Faraday, not physical objects. They are drawn by stepping from point to point in the direction the field vector points, with line density representing field strength. The field itself exists at *every* point in space — between the lines too. Where this document says "the $\mathbf{E}$ field points downward" or "the $\mathbf{B}$ field curls around the trace," it is shorthand for: the field vector at each point in that region has that direction and magnitude.

Two of Maxwell's Equations

As we will see, Maxwell's equations tell the full story in four lines, but the key insight is in two of them — Faraday's Law and the Ampère-Maxwell's Law — with Gauss's law providing the source-free divergence constraint.

$$\nabla \times \mathbf{B} = \underbrace{\cancel{\mu_0 \mathbf{J}}}_{\substack{\text{conduction}\\ \text{current}}} + \underbrace{\mu_0\, \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}}_{\substack{\text{displacement}\\ \text{current}}} \qquad \text{(Ampère-Maxwell)}$$

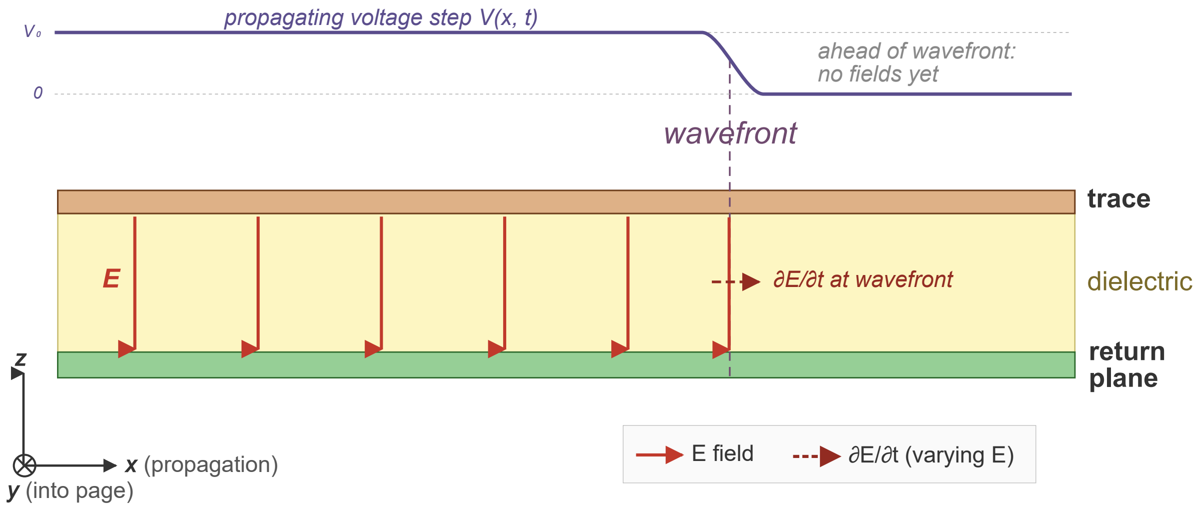
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \qquad \text{(Faraday)}$$

The **displacement current** term in the Ampère-Maxwell law is not a real current — no charge crosses the gap. A changing electric field acts as a source of magnetic field, just as a real current does. This is what allows $\mathbf{B}$ to be continuous across the copper-dielectric boundary, and it is what sustains the wave in the dielectric where $\mathbf{J} = 0$. In a material dielectric like FR-4, bound-charge polarization ($\partial \mathbf{P} / \partial t$) adds a second contribution, but the mechanism works in pure vacuum too.

Microstrip Geometry

Consider a signal trace running above a ground return plane, separated by a thin dielectric — the basic microstrip geometry of every PCB. For simplicity, this section treats the dielectric as the only medium between the conductors, ignoring the fringing fields that extend through the air above the trace. That simplification is revisited at the end of this section, where it gets a proper name.

In the source-free dielectric region ($\rho = 0, \mathbf{J} = \mathbf{0}$), the fields evolve as follows.



*Side view of microstrip **E** and **B** fields along the propagation direction.*

When a **voltage step** is applied at one end, the following chain of events unfolds:

1. The sudden voltage change creates an **electric field $\mathbf{E}$** between trace and return plane, pointing vertically (from trace down to return plane). This electric field exists only near the source end of the microstrip. $\mathbf{E}$ is called a vector field because it has an intensity and direction at every point in space. This field cannot remain localized.

2. The moment $\mathbf{E}$ appears, it is rising from zero — i.e. **changing in time**. Since the dielectric carries no conduction current ($\mathbf{J} = \mathbf{0}$), Ampère-Maxwell's Law reduces to the displacement-current term — a time-changing electric field $\mathbf{E}$ (RHS) **forces the magnetic field $\mathbf{B}$ to vary spatially** (LHS).

3. The $\mathbf{B}$ field created in step 2 is also rising from zero — i.e. **changing in time** (right side). According to Faraday's Law this time-changing magnetic field **forces the electric field $\mathbf{E}$ to vary spatially** (LHS) — extending $\mathbf{E}$ slightly ahead of where it began.

The electric field $\mathbf{E}$ points vertically (trace to return plane), but its magnitude changes as you move horizontally along the trace — field direction and variation are perpendicular. That is a wave front advancing.

Changing in time vs. changing in space. These two phrases carry the whole argument, so it is worth pausing on them.

- *Changing in time* ($\partial/\partial t$) — stand still at one point and watch the field rise, fall, or oscillate as the clock ticks. Units: [field] per second.
- *Changing in space* ($\nabla\times$) — freeze time and walk to a neighboring point; ask how the

value here differs from the value just over there. Units: [field] per meter.

Maxwell's curl equations link the two: a time change *here* forces a spatial difference *here*, which means the neighboring point has a different value, which forces *it* to change in time, and so on. A field that changed only in time would just pulse in place; one that changed only in space would be a frozen pattern. It is the coupling — time-derivative on one side, spatial-derivative on the other — that makes the disturbance *move*.

To summarize: once the electric field appears, a magnetic field arises alongside it. That changing magnetic field extends the electric field slightly ahead, which in turn extends the magnetic field further still. **Each field regenerates the other**. The wave is self-sustaining — it needs no electrons to carry it forward.

In other words

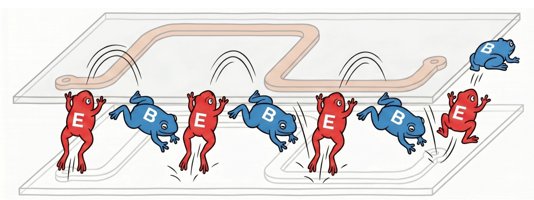
Here is how Feynman might have explained it. Chalk in one hand, no notes.

> Now look — you've got a copper trace, and underneath it a big sheet of copper called the return plane. Between them, a thin slab of plastic. That's it. That's the whole apparatus. And I want to tell you what happens when you flip a switch at one end and connect a battery.
> The instant you close the switch, there's a voltage between the trace and the plane. And whenever you have a voltage between two pieces of metal, there's an **electric field** between them. Bang — the field is just there, pointing from the trace down to the plane. Not in all of space, mind you — only right near the switch, because the rest of the trace hasn't heard the news yet.
>
> Now, here is where it gets interesting. The electric field went from zero to something. That's a change. And it turns out — and this is one of the most marvelous things in physics — that **a changing electric field makes a magnetic field**. Not "has a magnetic field associated with it." Makes one. Maxwell figured this out. The electric field changes in time, and a magnetic field curls up around it, right there in the plastic.
>
> OK, so now we have a magnetic field. And it is also going from zero to something. It's changing in time too. And Faraday — long before Maxwell — figured out the other half of the story: **a changing magnetic field makes an electric field**. So the magnetic field we just made, by changing, makes another electric field — a little bit further along the trace than the one we started with.

You see what's happening? The electric field made a magnetic field. The magnetic field made an electric field. The new electric field is further down the line. And now it is changing, so it makes another magnetic field, which makes another electric field, and off we go. The two fields are playing leapfrog[3], and they're heading down the trace at an enormous speed.

So when you flipped that switch — the light turned on because a little piece of light, essentially, rushed down the trace. Not the electrons. The field. The electrons are just sitting there wiggling; they're the audience, not the performers. The show is in the plastic, between the conductors, where the fields are doing their dance.

And that, really, is what every trace on every PCB is doing. It's not carrying electrons to some destination. It's guiding a little wave of light. Isn't that something?



Propagation

Steps 1–3 showed the first cycle. The same coupling, applied repeatedly, guarantees indefinite propagation.

As we have seen, in the source-free dielectric, the two laws simplify to:

$$\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \qquad \text{(Ampère-Maxwell)}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \qquad \text{(Faraday)}$$

These two equations couple time variation to spatial variation — and that coupling is what makes propagation inevitable. If $\mathbf{E}$ is changing in time at some point, the first equation forces $\mathbf{B}$ to have a spatial gradient there — so $\mathbf{B}$ at the neighboring point is different. At that neighboring point, $\mathbf{B}$ is now changing in time, and by the second equation this forces spatial variation in $\mathbf{E}$ — so $\mathbf{E}$ at the *next* point is different. And so on.

No mechanism "pushes" the wave forward. A time change here forces a spatial difference here, which means a different value there, which forces a time change there. The disturbance has no choice but to spread. The wave propagates because the mathematics forbid a localized disturbance from remaining localized.

How fast the electric and magnetic fields can build up is what sets the signal's propagation speed. The propagation and interaction of these fields is described by Maxwell’s Equations.

Expand if you ❤ math.
Take the curl of Faraday's law

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t}(\nabla \times \mathbf{B})$$

Substitute the simplified Ampère–Maxwell into the right side

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

Recall the vector identity

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} \qquad \text{(vector identity)}$$

Expand the left side using this vector identity

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

Apply Gauss's law:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \qquad \text{(Gauss's law)}$$

In the source-free dielectric there are no charges ($\rho = 0$), so this becomes $\nabla \cdot \mathbf{E} = 0$. The first term vanishes:

$$\nabla^2 \mathbf{E} = \underbrace{\mu_0 \, \varepsilon_0}_{1/v^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

Recognize the standard wave equation for any quantity propagating at speed v:

$$\nabla^2 f = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2} \qquad \text{(standard wave equation)}$$

Comparing the two, term by term, gives the speed of the wave propagation:

$$\frac{1}{v^2} = \mu_0\varepsilon_0 \quad \Rightarrow \quad v = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$$

Takeaway: combining Faraday and Ampère-Maxwell yields the standard wave equation with speed v. This speed of the EM wave follows directly from the constants for permeability (μ_0) and permittivity (ε_0):

$$v = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$$

The constants μ_0 and ε_0 follow from independent magnetic and electric experiments, respectively — neither involving light.

$$\left.\begin{aligned} \mu_0 &= 4\pi \times 10^{-7}\ \mathrm{H/m} \\ \varepsilon_0 &= 8.854 \times 10^{-12}\ \mathrm{F/m} \end{aligned}\right\}$$

Substituting the constants:

$$v = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.854 \times 10^{-12}}} \approx 299.8 \times 10^6\ \mathrm{m/s}$$

That is the speed of light c — derived entirely from electric and magnetic constants.

This was Maxwell's 1865 result. He started with two equations about how electric and magnetic fields change in space and time, combined them, and out fell the speed of light. That is one of the most remarkable results in all of physics.

To find the propagation speed (v') for a typical PCB dielectric glass epoxy (FR-4), we need to replace ε_0 with $\varepsilon_r\varepsilon_0$, where $\varepsilon_r \approx 4.2$.

$$v' = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 4.2 \times 8.854 \times 10^{-12}}} \approx 146.3 \times 10^6\ \mathrm{m/s}$$

The wave **propagation speed in FR-4** is therefore **~15 cm/ns**. To be precise, this is the bulk-FR-4 (or stripline) speed. On a microstrip, part of the field is in air above the trace, so the effective permittivity is a bit lower and the speed rises to ~17–18 cm/ns.

This is also the right moment to name the simplification we have been using. Throughout this section, $\mathbf{E}$ points vertically and $\mathbf{B}$ curls horizontally — both entirely perpendicular to the propagation direction. A wave mode where neither field has a longitudinal component is called **TEM** (Transverse Electromagnetic). A stripline, where the trace is sandwiched between two return planes in a uniform dielectric, supports a pure TEM mode. A microstrip does not: part of the field travels through the dielectric ($\varepsilon_r \approx 4.2$) and part through the air above ($\varepsilon_r = 1$). Because the two media have different propagation speeds, the fields cannot be purely transverse — small longitudinal components appear to satisfy the boundary conditions. The mode is called **quasi-TEM**. At the frequencies relevant to PCB design (up to a few GHz), the longitudinal components are small enough that the quasi-TEM approximation holds well, and the transmission-line model — characteristic impedance, propagation delay, reflections — remains valid.

In other words

Once more in Feynman's words:

> If you write down Faraday's and Maxwell's laws, and you do a little algebra (which I'll spare you), out pops a speed. And the speed is $1/\sqrt{\mu_0 \varepsilon_0}$, where μ_0 and ε_0 are just numbers you measured in a laboratory with magnets and charges, nothing to do with light at all. And when you plug them in, the speed is three hundred thousand kilometers per second. Which is the speed of light. Maxwell looked at this and said, my goodness, light is just this game of leapfrog between electric and magnetic fields.

1.2. Conductors as Waveguides

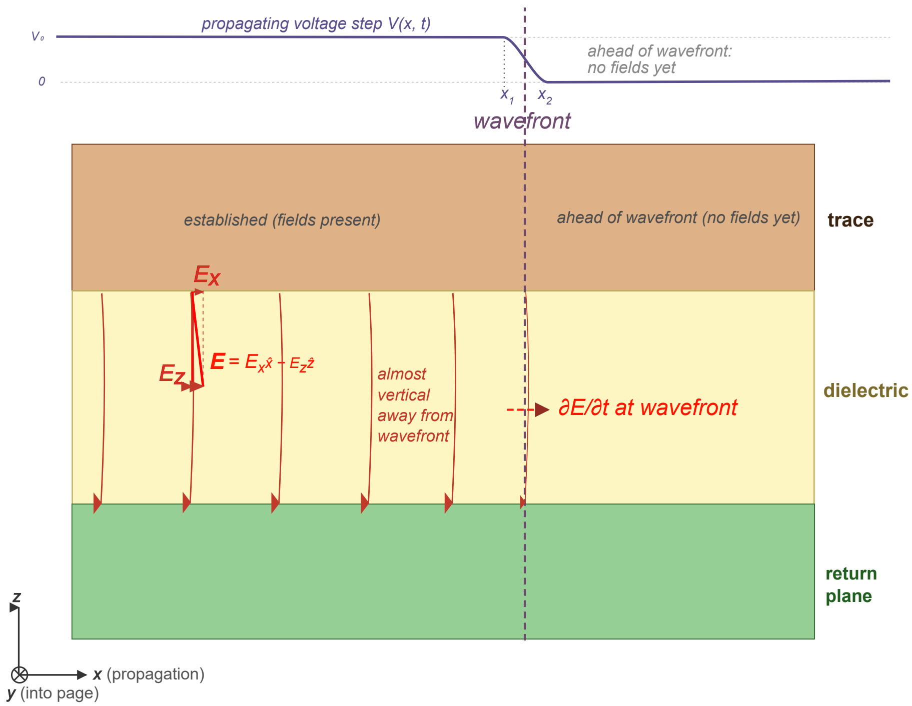
§1.1 looked at the dielectric — the wave, the energy, the propagation. Here we turn our attention to the trace and return plane. Inside the copper, there are free electrons, and they are not passive bystanders.

Components of the Electric Field

The observant reader might have noticed that the electric field between trace and return plane is not perfectly vertical — it tilts slightly, carrying a small horizontal component alongside the dominant vertical one. That tilt decomposes into two components:

$$\mathbf{E} = \hat{x}\,E_x - \hat{z}\,E_z$$

where $\hat{x}$ points along the trace (the propagation direction) and $-\hat{z}$ points from the trace to the return plane.



Microstrip electric field components.

The **free electrons** in the copper respond to each component differently:

- **Vertical component** E_z (dominant) — confines the wave to the dielectric by driving surface charges that cancel the field inside the metal.
- **Horizontal component** E_x — arises from two distinct causes depending on where along the line you look:
  - **At the wavefront** (strong) — the voltage transitions from signal level to zero over a short distance, producing a steep spatial gradient ($E_x = -\partial V/\partial x$). This initiates the current.
  - **Behind the wavefront** (small) — the finite resistance of the copper causes a gradual voltage drop with distance, producing a gentler E_x that sustains the current against resistive drag.

The following subsections unpack each component in detail, starting with the vertical.

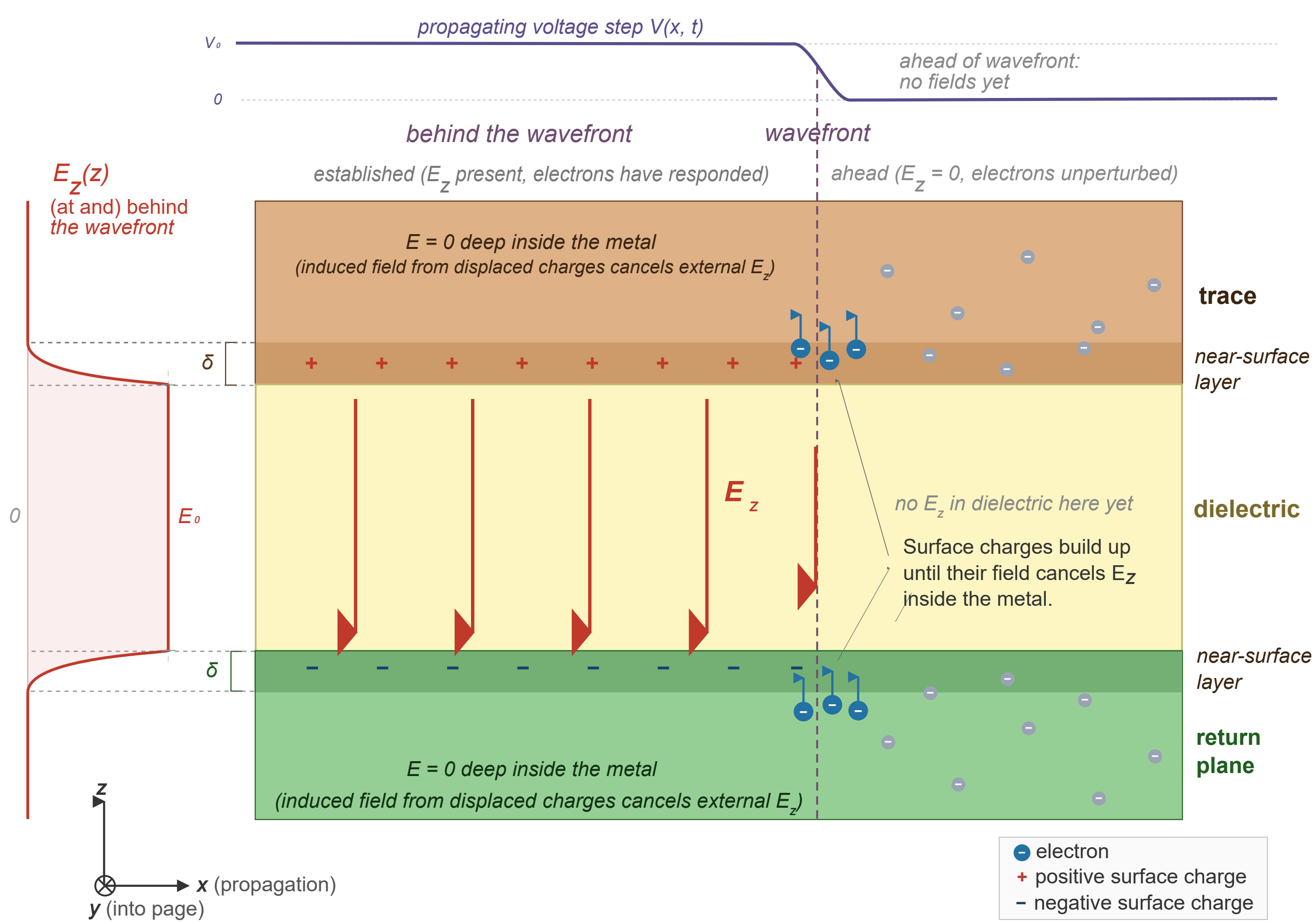
Vertical Component E_z

As the wave front reaches a section of the conductor, E_z there rises from zero to some value. The force on an electron is $\mathbf{F} = (-e)\,\mathbf{E}$ (where $e > 0$). With E_z pointing from trace down to return plane, electrons in *both* conductors are pushed *upward*:

- in the trace, they move away from the dielectric-facing surface, leaving it positively charged;
- in the return plane, they move toward the dielectric-facing surface, making it negatively charged.

The local redistribution of charge caused by E_z charges the distributed capacitance of the line.

The result is a pair of opposite surface charges that act like a parallel-plate capacitor. Inside each conductor, the field from the nearby surface charge opposes the incoming E_z — pointing upward in the trace (away from the positive surface) and downward in the return plane (away from the negative surface). The electrons keep moving until their self-generated field exactly cancels the incoming field, driving the net $\mathbf{E}$ inside the metal to zero.



Microstrip E_z confinement.

The cancellation happens almost instantly. What little field penetrates the metal decays within one skin depth $\delta = \sqrt{2/(\omega\mu\sigma)}$ (~66 µm at 1 MHz, ~2 µm at 1 GHz).

Because the field cannot extend into the copper, it is forced to remain in the dielectric between the two conductors. This is what **confines the wave**: without the electron response, the field would radiate away instead of propagating along the line.

In other words

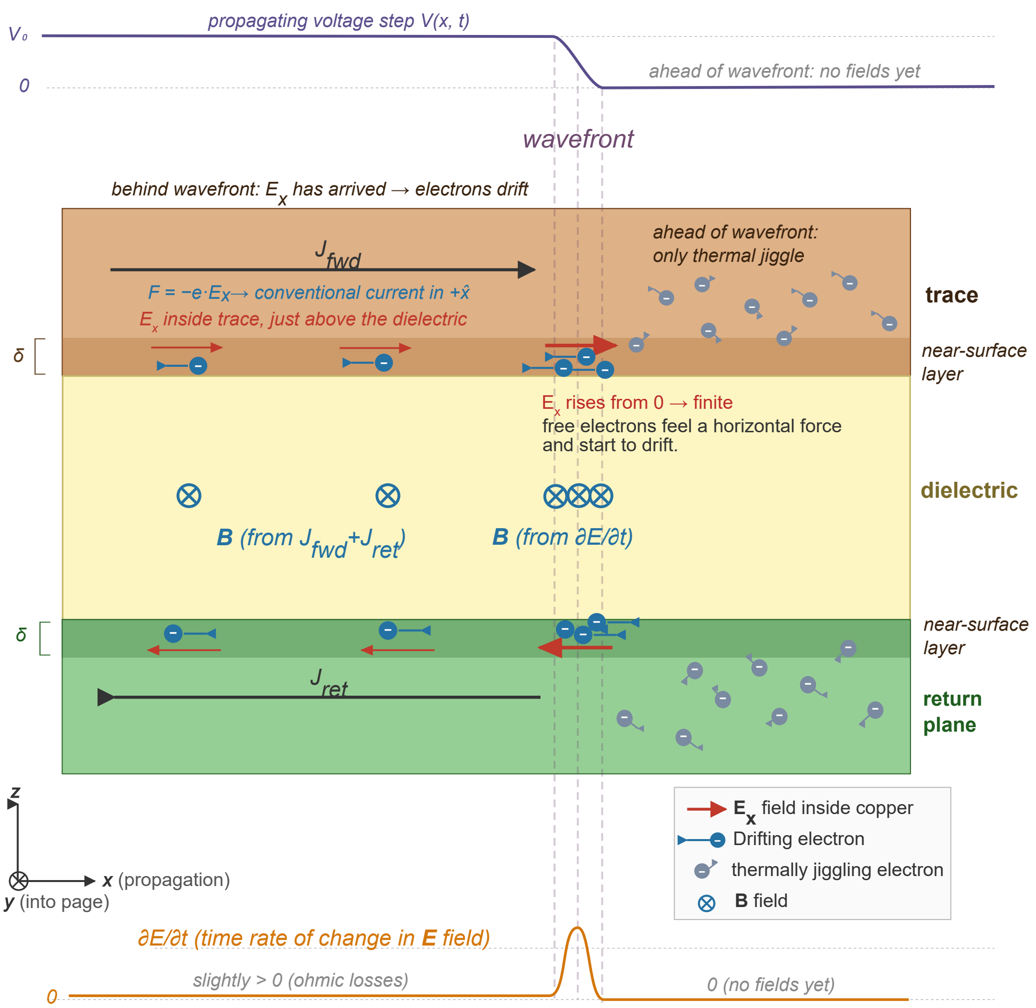
> Here's what the down-field does — and I think this is the most important thing happening on the whole PCB.
>
> The wave comes along, and it's got this big electric field pointing from the trace down to the return plane. Now, inside the copper there are free electrons — trillions of them, just sitting there. And that field grabs them. In the trace, it yanks them *upward*, away from the bottom surface. In the return plane, same thing — it shoves them *up* toward the top surface. So now you've got a deficit of electrons on the bottom of the trace — that's a positive surface charge — and a pileup of electrons on the top of the return plane — that's a negative surface charge. You've made a little capacitor, right there, between the two conductors.
>
> But here's the thing. Those piled-up charges have their own electric field. And it points the *other way* — from the positive surface up toward the negative surface. So inside the copper, you've now got two fields fighting: the incoming field from the wave, pushing down, and the self-field from the surface charges, pushing up. And the electrons keep rearranging themselves until those two fields *exactly cancel*. The net field inside the metal goes to zero.
>
> How far does the incoming field get before the electrons shut it down? Almost nowhere. At a gigahertz, the field penetrates about 2 microns — two millionths of a meter — before the electrons have killed it. That's the skin depth. The copper in a PCB trace is tens of microns thick. The field doesn't stand a chance.

Horizontal Spatial Gradient

At the wavefront, the voltage rises from zero to signal level over a short distance (as seen at a fixed point when the wave arrives), while a little further ahead it is still zero. This steep spatial gradient is a horizontal electric field:

$$E_x = -\frac{\partial V}{\partial x}$$

The free electrons — previously in thermal equilibrium with no net motion — feel a force $F_x = (-e)\,E_x$ and begin to move horizontally. The **current is the electrons' response to the electric field**.



Horizontal current arising from sequential vertical charge displacement at the wavefront.

In other words

> Now, what about the along-field — the dotted horizontal arrow? Where does that come from?

It comes from the wave *going somewhere*. The voltage on the trace right here is at signal level. But a millimeter ahead, the wave hasn't arrived yet — the voltage is still zero. So you've got a voltage that changes from one place to another. And wherever the voltage changes from place to place, there's an electric field pointing from high to low. That's the along-field. It's a spatial gradient — $E_x = -\partial V/\partial x$ — and right at the wavefront, where the voltage drops from full to zero in a very short distance, it's steep.

The free electrons in the copper feel that field and start to drift. Very slowly, mind you — but they all drift together, and that is what we call a **current**. The current didn't cause anything. The field caused the current. The field shows up first, the electrons react.

Horizontal Component E_x

So far we have described how E_x arises at the wavefront itself. But the conductor is resistive, and that changes the picture behind the wavefront as well.

The drifting electrons scatter off nuclei, converting drift kinetic energy to lattice vibrations (heat). This is a drag force opposing the current. If nothing compensated for this drag, the current would decay. But the current can't just decay — the wavefront is still advancing, still demanding current to charge the next section. So the system has to sustain it. Here's how:

The scattering drains energy from the wave. That means the wave amplitude (voltage) drops slightly with distance along the line. A voltage that decreases with x means there's a spatial gradient — the same relation as before, now in a different context:

$$E_x = -\frac{\partial V}{\partial x}$$

This E_x is what re-accelerates the electrons against the drag.

The resistance is the root cause. E_x is the system's response to resistance — the "cost" of pushing current through a lossy conductor. On a superconducting line (no scattering), there would be no drag, no voltage drop, no E_x, and no loss.

Behind the wavefront, the surface charge at any given section is already established. But current still flows through those sections — not to charge them, but to supply the wavefront that's still advancing ahead.

Think about what happens at the wavefront right now, at section x_3:

- E_z pushes electrons upward in the trace → bottom surface becomes positive (electron deficit).
- Those displaced electrons don't pile up at the top — they flow back through the conductor toward the source
- That flow of electrons through all the sections behind the wavefront is J_{fwd}.

So each already-charged section behind the wavefront is acting as a conduit. Its own surface charges are set, but current passes through it to feed the front. It's like a pipe that's already full of water — water still flows through it to supply whatever is at the end.

The moment the wavefront stops advancing (reaches a matched load), the current doesn't vanish — it continues, now supplying the load instead of charging new sections.

This also clarifies why "surface charge established" and "current flowing" aren't contradictory. In any DC circuit, the wire has stable surface charges (they guide the current), yet current flows through it continuously. The transmission line behind the wavefront is in exactly that state — a quasi-DC condition where the surface charges steer the current, and E_x sustains it against resistive drag.

In other words

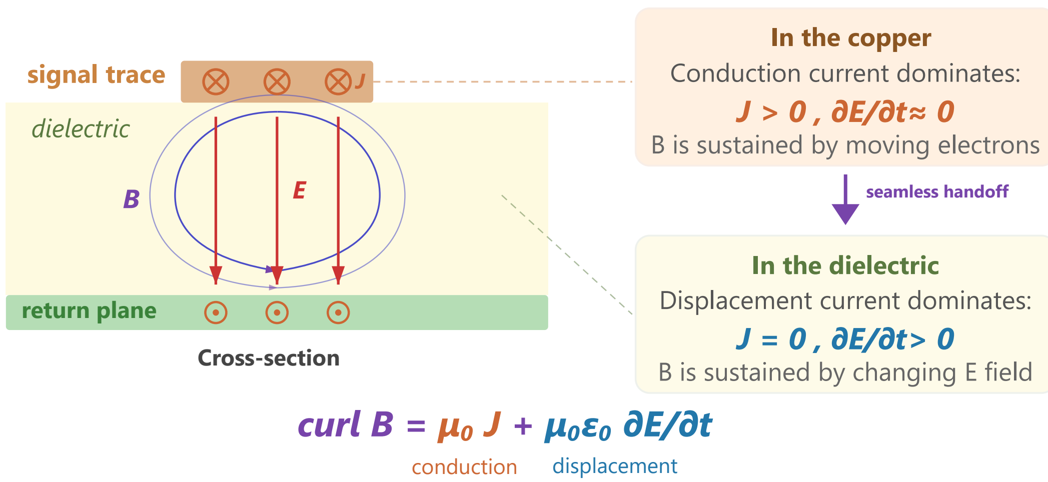
> The electrons don't drift for free. They bang into copper atoms as they go — that's resistance — and every collision turns a little bit of drift energy into heat. That friction drains energy from the wave, so the wave amplitude drops slightly with distance. And a voltage that drops with distance means — there it is again — a spatial gradient. A gentler one this time, not the steep cliff at the wavefront, but enough to keep pushing the electrons forward against the drag. Behind the wavefront, the along-field is the system's way of paying the resistive tax.
>
> And what about the sections the wavefront has already passed? Their surface charges are set — the capacitor is charged. But current still flows through them, because the wavefront is still advancing ahead, still demanding current to charge the *next* section. It's like a pipe that's already full of water — water still flows through it to supply whatever is at the end. The moment the wavefront reaches a matched load, the current doesn't vanish — it just switches from charging new sections to supplying the load.

Putting It Together

> So put it all together: - The wave is in the plastic. - The *down-field* from the wave shoves electrons around on the copper surfaces. Those electrons build a wall that keeps the wave trapped in the plastic. - The *along-field* from the wave pushes other electrons along the trace, giving you the little current you measure with your ammeter.
> The wave is the boss. The electrons are the help. And what the electrons do — the thing we call "current" — is not how the energy gets from here to there. The energy is out in the dielectric, in the fields. The current is just the electrons reacting to the field, the same way a line of dominoes reacts to the first push. The dominoes aren't transporting the energy down the line — the *falling pattern* is. And on a PCB, the falling pattern is the field.
>
> And that's really all there is to it. The hard part is believing it.

1.3. Sources of the Magnetic Field

§1.1 showed that a changing $\mathbf{E}$ field in the dielectric creates a $\mathbf{B}$ field (displacement current). §1.2 showed that the wave drives a conduction current $\mathbf{J}$ in the copper — and from introductory physics we know that a current also creates a $\mathbf{B}$ field. That raises an obvious question: are there two competing magnetic fields on a microstrip, one from the wave and one from the current? This section shows there is only one — and that the two sources are just two terms in the same equation.



Cross-section showing one continuous B field sustained by two sources: conduction current in the copper and displacement current in the dielectric.

The horizontal current $\mathbf{J}$ in the trace creates a $\mathbf{B}$ field curling around it. A natural question: is this a separate magnetic field competing with the wave's own $\mathbf{B}$? No — it is the *same* field. Ampère–Maxwell makes this explicit:

$$\nabla \times \mathbf{B} = \underbrace{\mu_0 \, \mathbf{J}}_{\substack{\text{conduction current} \\ \text{in the copper}}} + \underbrace{\mu_0 \, \varepsilon_0 \, \frac{\partial \mathbf{E}}{\partial t}}_{\substack{\text{displacement current} \\ \text{in the dielectric}}}$$

There is one continuous $\mathbf{B}$ field, but it has two sources depending on where you are. Inside the copper, the conduction term ($\mu_0 \mathbf{J}$) dominates — free electrons are moving, and their motion sustains $\mathbf{B}$. In the dielectric, there are no free electrons ($\mathbf{J} = 0$), so the displacement term takes over — the changing $\mathbf{E}$ field sustains $\mathbf{B}$ just the same. At the copper-dielectric boundary, the two sources hand off seamlessly. The $\mathbf{B}$ field does not care which term is producing it; it is one smooth, continuous solution across the interface.

In other words

Imagine Feynman is still at the chalkboard. Someone in the third row raises a hand.

> *"Professor, I'm confused. You just told us the wave has a magnetic field around it — that was part of the leapfrog thing. But now you're saying the electrons in the copper are flowing, and a flowing current also makes a magnetic field. So which is it? Are there two magnetic fields here, or what?"*
>
> That's a wonderful question. And the answer — which I think is one of the most beautiful

things Maxwell ever did — is that there's only **one** magnetic field. There's just one field in space, wrapping around the trace. But that field has *two different things that can keep it going*, and which one is keeping it going depends on *where you are*.

Let me show you. Maxwell's Ampère law — the real one, with his addition — says the curl of $\mathbf{B}$ is two things added together:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

That first piece — $\mu_0 \mathbf{J}$ — is just the old-fashioned Ampère's law. Current flowing makes magnetic field. Every kid who ever wrapped wire around a nail knows this.

That second piece was Maxwell's great invention. He realized Ampère's law couldn't be complete because if you had a capacitor, the current comes *into* one plate and *out of* the other, but in between — in the empty space — there's no current flowing. And yet the magnetic field has to be continuous; it can't just stop at the plate and start up again on the other side. Something has to keep it going across the gap. And Maxwell said: the thing that keeps it going is the *changing electric field* between the plates. He called it the **displacement current**, even though nothing is really being displaced. It's just a changing field that *looks like* a current, from the magnetic field's point of view.

Now look where we are on this PCB. You've got copper up top, copper down below, and plastic in between.

Inside the copper, electrons are flowing. So $\mathbf{J}$ is big. And the electric field in the copper is essentially zero — we just spent the last lecture explaining why. So the second term is nothing. The $\mu_0 \mathbf{J}$ piece is doing all the work.

Inside the plastic, there are no free electrons. $\mathbf{J} = 0$. Nothing is flowing through the dielectric. So the first term is nothing. But the electric field is enormous in there, and it's changing in time as the wave goes by. So the second term — the displacement current — is doing all the work.

And here's what I want you to appreciate. You walk from the plastic up into the copper, crossing the boundary, and the $\mathbf{B}$ field doesn't care. It doesn't notice. It's perfectly smooth. It has the same value just below the surface as just above. What changes is *who's responsible for it*. Down in the plastic, a changing electric field is keeping $\mathbf{B}$ alive. Up in the copper, moving electrons are keeping it alive. The field itself is completely oblivious — it just wraps around the trace as one continuous tube of magnetism.

That was Maxwell's real insight. Not that current makes magnetism — we knew that. Not that changing fields make fields — Faraday had half of that. Maxwell's piece was realizing that *a changing electric field is, for the purposes of magnetism, every bit as good as a current.* Nature doesn't care which one is feeding the field. It'll take either. And on a PCB, inside the copper it takes one, inside the plastic it takes the other, and the result is a seamless, beautiful, continuous magnetic field wrapping the whole transmission line.

One System, Two Views

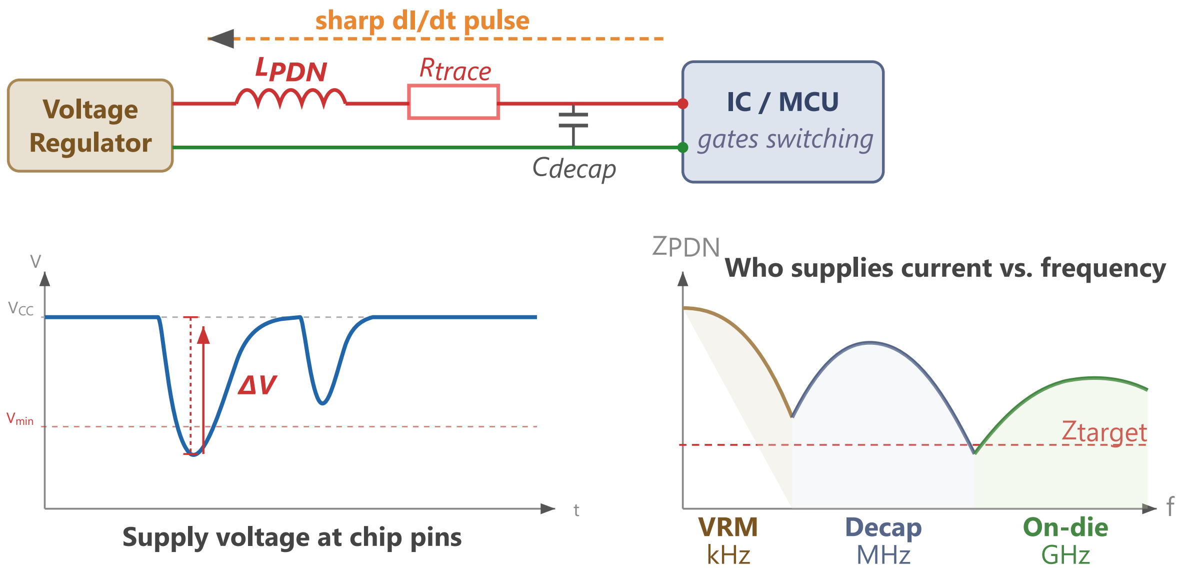
It is tempting to think of "the wave in the dielectric" and "the current in the copper" as two separate things that happen to coexist. They are not. They are two views of a single electromagnetic solution, and neither can exist without the other.

- Without the surface-charge redistribution, the boundary condition that cancels $\mathbf{E}$ inside the metal would not be satisfied. The field would not be confined. There would be no guided wave — just radiation.
- Without the propagating field, there would be nothing to drive the electrons. No E_x means no $\mathbf{J}$. The current would not exist.

The field drives the current. The current shapes the field. They are mutually dependent — one self-consistent system, seen from different sides of the copper surface.

So far, we have followed one signal on one trace — its wave, its return current, its confinement. But every chip on the board also needs a stable supply voltage, delivered through its own set of traces, vias, and planes. Those power paths are transmission lines too, and they are subject to the same field physics. When the current through them changes abruptly, the results are not subtle.

1.4. Rail Collapse



PDN inductance causes voltage sag when current changes sharply. The VRM, decoupling capacitors, and on-die capacitance each cover a different frequency band.

The power and ground paths that feed every chip on the board are themselves transmission lines with impedance — and when the current through them changes, that impedance produces voltage noise.

Every time a chip switches its outputs or its internal gates toggle, it draws a sharp pulse of current from the power rail. That current passes through the inductance L_{PDN} of the power distribution network (PDN) — the planes, traces, vias, and decoupling capacitors between the voltage regulator and the chip. The resulting voltage drop is:

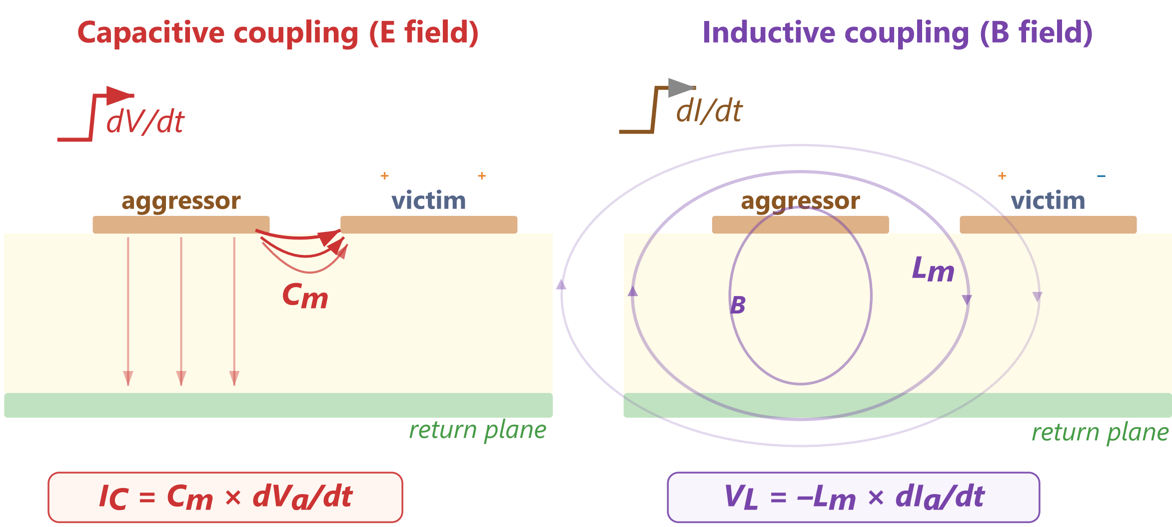
$$\Delta V = L_{\text{PDN}} \times \frac{dI}{dt}$$

This is trace and via inductance resisting sudden changes in current. The chip sees its supply rail sag momentarily, reducing the voltage between its power and ground pins. If the sag is large enough, the chip misinterprets logic levels or produces timing errors. The design goal is to minimize L_{PDN} across the full frequency range over which the chip draws current — the details are covered in §2.1.

1.5. Crosstalk

Crosstalk is what happens when the EM field of one trace overlaps with the EM field of another. In circuit theory we talk about parasitic capacitance and mutual inductance as if they are discrete components accidentally added to the circuit. In field theory the picture is more honest: there is one EM field in the dielectric, and that field does not know which trace "owns" it. If two traces share the same dielectric volume, their fields overlap, and the overlap is the crosstalk. C_m and L_m are just circuit-theory approximations of that field overlap.

There are two coupling mechanisms — capacitive and inductive — and both are direct consequences of Maxwell's equations.



Capacitive coupling (E field fringing between traces) and inductive coupling (B field threading the victim loop), with near-end and far-end polarity summary.

Capacitive (electric field)

The $\mathbf{E}$ field from a signal trace does not terminate exclusively on its own return plane. Some field lines — especially the fringing fields at the edges of the trace — terminate on nearby conductors instead: an adjacent trace, a via, a component pad.

When the aggressor trace changes voltage, its $\mathbf{E}$ field changes. That changing field induces a displacement current ($\varepsilon_0 \varepsilon_r \frac{\partial \mathbf{E}}{\partial t}$) onto the victim trace — depositing charge on it, just as it would on a capacitor plate. The victim trace sees a current spike proportional to the rate of change of the aggressor's voltage V_a:

$$I_C = C_m \frac{dV_a}{dt}$$

where C_m is the mutual parasitic capacitance between the two traces — determined by their overlap area, separation distance, and the dielectric constant of the material between them. Closer traces and longer parallel runs increase the coupling. A higher ε_r increases mutual capacitance, but also tightens field confinement to the return plane — the net effect depends on geometry.

This is purely Gauss's law at work: electric field lines must terminate on a conductor, and if another trace is closer or more convenient than the return plane, some of them will land there instead.

Inductive (magnetic field)

The current in the aggressor trace creates a $\mathbf{B}$ field curling around it. That field extends into the surrounding space — including through the loop formed by the victim trace and its return plane. When the aggressor's current I_a changes, the $\mathbf{B}$ field through the victim's loop changes. Faraday's law says a changing magnetic flux through a loop induces a circulating electric field — which drives a voltage:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{Faraday}$$

The induced voltage V_L is proportional to the rate of change of the aggressor's current:

$$V_L = -L_m \frac{dI_a}{dt}$$

where L_m is the mutual inductance between the two trace-return-plane loops. It depends on how much of the aggressor's $\mathbf{B}$ field threads through the victim's loop — set by the physical distance between traces, the height above the return plane, and the length of the parallel run.

Inductive crosstalk is worst where the return path is constrained. On a PCB with a continuous return plane, the return current mirrors directly under the trace, keeping the loop area small. But through connectors, packages, and vias, multiple signals often share a single return pin instead of a wide plane. The return currents are forced through a common impedance, the loop areas grow, and the mutual inductance between aggressor and victim increases sharply.

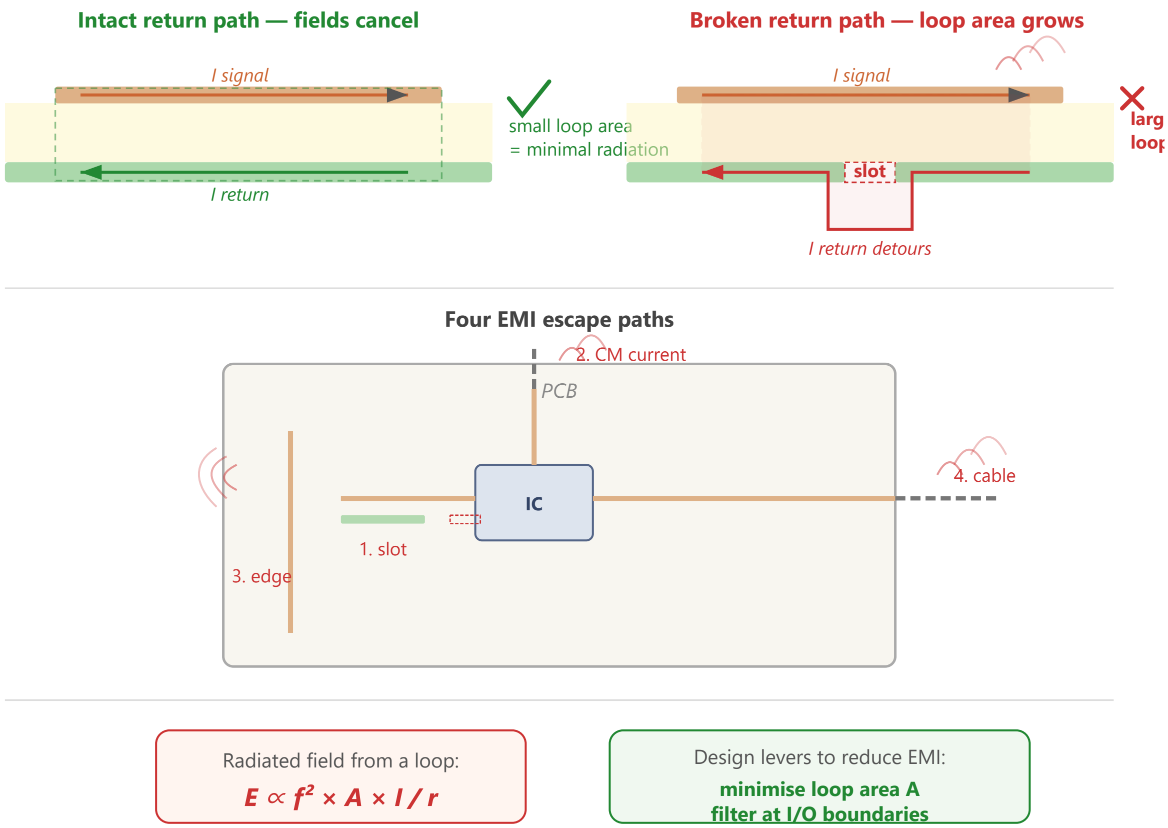
Both Matter

The capacitive and inductive coupled signals arrive at the victim differently:

- **Capacitive coupling** injects current into the victim trace. That current splits and travels in both directions — toward the near end and the far end. Both ends see a pulse of the same polarity as the aggressor.
- **Inductive coupling** induces a current that opposes the aggressor's current (Lenz's law). That current flows toward the near end, producing a pulse of the same polarity as the aggressor there, and a pulse of opposite polarity at the far end.

At the **near end** (closest to the aggressor's source), the capacitive and inductive components have the same polarity — they add. Near-end crosstalk (NEXT) is therefore the larger of the two. At the **far end**, they have opposite polarity — they partially cancel. In a stripline (homogeneous dielectric), the cancellation is exact and far-end crosstalk (FEXT) is zero. In a microstrip, the even- and odd-mode velocities differ slightly, so FEXT is nonzero but typically smaller than NEXT.

1.6. Electromagnetic Interference



Top: intact return path (fields cancel, minimal radiation) vs. broken return path (enlarged loop radiates). Middle: four EMI escape paths on a PCB. Bottom: radiation formula and design levers.

Sections §1.1 through §1.5 describe what happens when the EM field stays where it belongs — confined between a trace and its return plane. EMI is what happens when it escapes.

A current loop is an antenna. For a small loop (perimeter $\ll$ wavelength), the radiated electric field at distance r is proportional to the square of the frequency:[4]

$$E \;\propto\; \frac{f^2\, A\, I}{r}$$

Every signal on the board has a forward path (the trace) and a return path (the current in the return plane directly beneath it). When both paths are intact and close together, the loop area is tiny — just the trace length times the dielectric thickness. The fields from the forward and return currents are equal and opposite, and they cancel at a distance. Almost no energy escapes.

EMI appears when that cancellation breaks down:

- **Broken return path.** A slot in the return plane, a missing via at a layer transition, or a signal crossing between power islands forces the return current to detour. The loop area grows — and since radiated power scales with A^2, even a modest detour has outsized consequences.
- **Common-mode currents.** If the return current cannot mirror the signal current exactly — because of an asymmetry, a ground impedance, or a cable acting as a second antenna — the imbalance becomes a common-mode current. Common-mode currents flow on the outside of cables and along board edges, where there is no equal-and-opposite field to cancel them. Even a few microamps of common-mode current at VHF frequencies can exceed emission limits.
- **Board-edge fringing.** The EM field guided by a microstrip trace extends laterally beyond the trace edges. If a high-speed trace runs near the board perimeter, those fringing fields reach the edge of the return plane and radiate — there is no copper beyond the edge to contain them.
- **Connector and cable radiation.** Every conductor that leaves the board — a power cable, a sensor wire, a USB connection — is a potential antenna. The board's internal switching noise couples onto the cable as common-mode current, and the cable radiates it. This is typically the dominant EMI path in a system like OPNhydro, where multiple cables connect to off-board sensors and motors.

The physics comes from the same Maxwell's equations as §1.1. The difference is context: signal integrity asks whether the field arrives at the receiver correctly; EMI asks whether the field arrives somewhere it should not.

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2. From Physics to Layout

Chapter 1 established how signals actually travel on a PCB — as EM waves guided by copper boundaries, with energy in the dielectric and currents as the electrons' response to the field. This chapter turns that physics into layout decisions.

The sections proceed from rules to physical build to final strategy:

- **§2.1** distills the field theory into layout rules covering signal quality, crosstalk, rail collapse, and EMI.
- **§2.2** explains the OPNhydro stack-up — a Low-EMI four-layer arrangement driven by the 6.5 A peak on the 24 V input rail and the isolation moats around the pH and EC islands.
- **§2.3** specifies the dielectric and copper.
- **§2.4** separates the noisy motor-control, digital, and sensitive analog domains.
- **§2.5** covers enclosure and mechanical constraints.
- **§2.6** derives trace widths from current, impedance, and thermal limits.
- **§2.7** assembles all of the above into a concrete layout strategy.

Each rule in this chapter ties back to a specific result from Chapter 1. The aim is that no rule is a folklore prescription — every one has a physical reason behind it.

2.1. PCB Design Rules

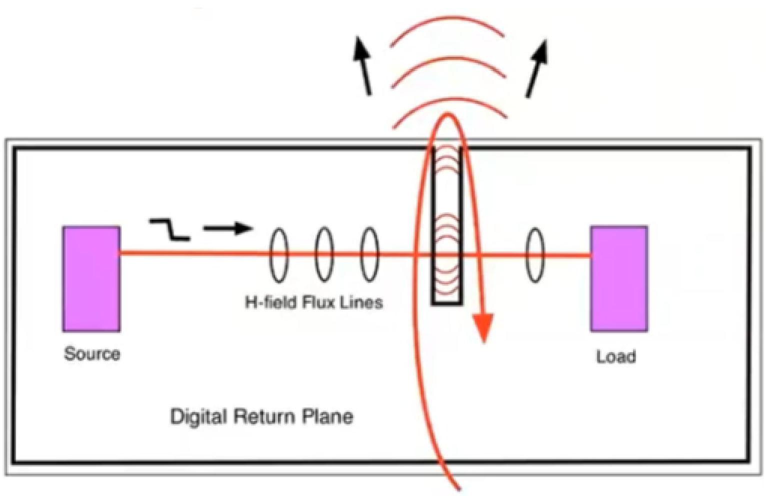
Everything in §1.1 through §1.5 leads to a single conclusion: the signal energy travels as an EM wave through the dielectric, guided by the copper boundaries. The trace is one wall, the return plane is the other. The copper confines the field (E_z cancellation), the dielectric carries it forward (displacement current), and the return current in the return plane provides the equal-and-opposite $\mathbf{B}$ that prevents radiation (field cancellation at a distance).

When that field structure breaks down, the consequences fall into four categories: degraded signal quality on a single net (reflections, ringing), rail collapse in the power distribution network (§1.3), crosstalk between adjacent nets (§1.4), and radiated EMI (§1.5). Every PCB layout rule exists to prevent one or more of these — by keeping the field confined, the return path intact, and the coupling between unrelated fields to a minimum.

Signal Quality

A signal traveling along a trace is an EM wave guided by the trace and its return plane. Anything that disrupts the wave's propagation — an impedance discontinuity, a missing return path, a stub — causes part of the energy to reflect back toward the source. The reflected wave interferes with the forward wave, producing ringing, overshoot, and timing uncertainty on the net.

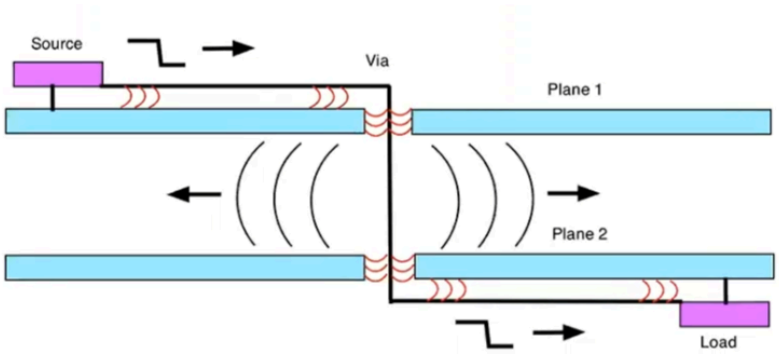
Rule 1a — Maintain a continuous return plane. The $\mathbf{B}$ field from the forward current in the trace and the return current in the return plane are equal and opposite. They cancel at a distance, keeping the energy confined. Gauss's law for magnetism ($\nabla \cdot \mathbf{B} = 0$) requires magnetic field lines to close: with a continuous plane, they close tightly. Interrupt the plane — a slot, a cutout, a missing pour — and the loop area grows, the impedance changes, and the wave partially reflects.



Trace crossing a gap in the return plane.
(Courtesy: Kenneth Wyatt)

> "Forget the word ground. Every signal has a return path. Think return path and you will train your intuition to look for and treat the return path as carefully as you treat the signal path." -- Eric Bogatin

Rule 1b — Provide return vias at layer transitions. When a trace passes through a via, the EM wave transfers between layers. The wave is not just the trace — it is the field between the trace and its reference plane. If the return plane changes (say, from L2 to L3), the return current must also transition. Without a nearby ground via, the return path detours, the loop area grows, and the wave leaks between the return planes — causing both reflections on the signal net and interference with other signals in that space.



Trace passing through two planes with via.
(Courtesy: Kenneth Wyatt)

If the planes are the same potential, prevent leakage with nearby stitching vias between them. If they are different potentials, place stitching capacitors as close to the signal via as possible.

Crosstalk

Both coupling mechanisms — capacitive (C_m, from overlapping $\mathbf{E}$ fields) and inductive (L_m, from overlapping $\mathbf{B}$ fields) — depend on how much of the aggressor's field volume overlaps with the victim's. Every mitigation strategy reduces that overlap.

Rule 2a — Increase trace spacing. The fringing $\mathbf{E}$ field that causes capacitive coupling falls off roughly as $1/d$ with distance. The $\mathbf{B}$ field that causes inductive coupling follows Ampere's Law and falls off as $1/d$. The 3W rule (center-to-center spacing of at least 3× the trace width) is a practical approximation of this. For critical traces (analog sensors, clocks), use $5W$.

Rule 2b — Minimize parallel run length. Both C_m and L_m are proportional to the length over which two traces run in parallel. The coupling is cumulative — every millimeter of shared dielectric adds to the total. Where two sensitive traces must be routed near each other, cross them at 90° rather than running them in parallel.

Rule 2c — Reduce trace height above the return plane. The closer a trace is to its return plane, the more tightly the $\mathbf{E}$ and $\mathbf{B}$ fields are confined directly underneath. Less field energy spills sideways into the neighboring trace's volume.

Rule 2d — Interpose a return plane between signal layers. A grounded conductor between two signal layers terminates $\mathbf{E}$ field lines from traces above (Gauss's law — the lines land on the return plane instead of reaching the layer below) and provides a local return path that contains the $\mathbf{B}$ field, blocking inter-layer coupling.

Rule 2e — Separate functional domains. Motor control traces and analog sensor traces must not share the same dielectric space. Keep traces on adjacent layers perpendicular to each other to minimize the parallel run length between layers.

Rail Collapse

Every time a chip switches, it draws a sharp current pulse from the power rail. That pulse passes through the inductance of the power distribution network, producing a voltage drop $\Delta V = L_{\mathrm{PDN}} \times \frac{dI}{dt}$. The goal is to minimize the PDN inductance across the full frequency range over which the chip draws current.

Rule 3a — Tightly couple power and return planes. A power plane and return plane separated by a thin dielectric (2–3 mil) form a parallel-plate capacitor with very low inductance. This provides broadband decoupling across the entire board area — the EM field between the planes can supply current before the discrete capacitors or the regulator can respond. This design uses two GND planes (L2, L3) with power routed as traces rather than a dedicated plane, so broadband plane decoupling is achieved through discrete capacitors instead (see §2.2).

Rule 3b — Use multiple, low-inductance decoupling capacitors. A single capacitor has parasitic lead and via inductance that limits its effectiveness above its self-resonant frequency. Multiple smaller capacitors in parallel reduce the effective inductance (inductances in parallel divide). Place them as close to the chip's power pins as physically possible — every millimeter of trace adds inductance.

Rule 3c — Minimize power and ground lead length in packages. The inductance of the bond wires and package leads between die and PCB is often the dominant contributor to L_{PDN} at high frequencies. Packages with multiple, short power and ground pins (QFN, BGA) have lower inductance than those with long leads (SOIC, DIP).

Rule 3d — Rely on on-chip decoupling for the highest frequencies. Above ~100 MHz, no external capacitor can respond fast enough — the path inductance from capacitor to die is too high. Modern ICs include on-die decoupling for this reason. The PCB designer's job is to keep L_{PDN} low at the frequencies below that.

EMI — Radiated Emissions

EMI is not a separate problem — it is the consequence of every other problem listed above. When a return path is broken, the $\mathbf{B}$ fields stop cancelling and the loop radiates. When crosstalk couples energy onto an unintended trace, that trace becomes an unintentional antenna. When a rail collapses, the transient current loop radiates at the switching frequency and its harmonics.

Rule 4a — Minimize loop area. Every current — signal, power, return — forms a loop. The radiated power from a loop is proportional to the loop area squared and to the fourth power of frequency: $P_{\mathrm{rad}} \propto A^2 f^4$. Keep traces close to their return planes. Use ground vias at every layer transition. Route power close to its return.

Rule 4b — Contain the fields at board edges. EM fields that reach the edge of the PCB can radiate freely — there is no conductor to confine them. Pull traces and pours back from the board edge by at least 20× the dielectric thickness (the 20H rule). Place return plane stitching vias along the board perimeter to create a continuous shield.

Rule 4c — Filter at I/O boundaries. Every cable attached to the board is a potential antenna. Place filtering (ferrite beads, capacitors, common-mode chokes) at the point where signals enter or leave the board, before the field has a chance to propagate onto the cable.

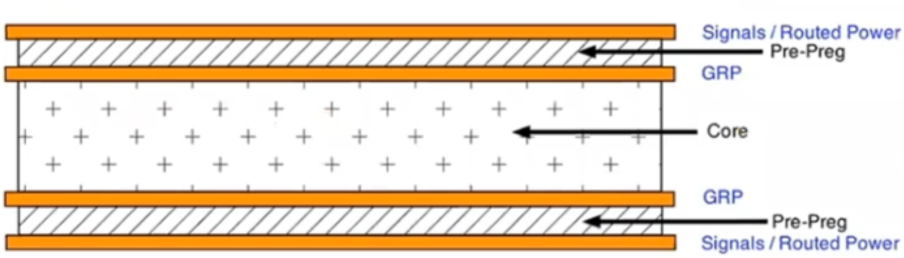
2.2. PCB Stack-up

The PCB has two hard constraints that drive most of the other design decisions. First, the 6.5 A peak current on the 24 V input rail requires copper heavy enough to handle that current without excessive resistive heating. Second, the isolation moats around the pH and EC islands must be maintained through all four layers, which means the layer stack-up cannot be an afterthought.

The **typical 4-layer stack-up** is SIG/GND/PWR/SIG. This design does not use it for two reasons. First, the power and return planes would be separated by the full core distance — too far apart for effective high-frequency decoupling (2–3 mil max is needed). Second, signals on Layer 4 would be referenced to the power plane rather than GND, which only works if the power and return planes are tightly coupled with adequate decoupling capacitors.

Instead, we opt for the **Low EMI 4-layer stack-up** as shown below.

Layer	Name	Function	Components
L1	Top	Sensitive signals / routed power	ESP32, LiDAR, I2C, UART, EZO, BNC, 3V3/5V power traces
L2	GND	Ground return plane	Primary return reference for L1 signals
L3	GND	Ground return plane	Primary return reference for L4 signals
L4	Bottom	Noisy signals / routed 24V power	Stepper drivers, MOSFETs, 24V power traces



Lower EMI in 4-layer PCB.
(Courtesy: Kenneth Wyatt)

2.3. PCB Materials

The design specifies a **4-layer PCB with 2 oz copper on the outer layers**. The heavier copper on L1 and L4 keeps resistance and heat low on the high-current 24V traces. The two inner layers (L2 and L3) use standard 1 oz copper, which is sufficient for the return planes they carry.

PCB finish: HASL (Hot Air Solder Leveling) is sufficient and lowest cost for the packages used in this design (SSOP-20 at 0.65 mm pitch and larger). ENIG (Electroless Nickel Immersion Gold) is only worth the premium if a future revision adds true fine-pitch parts (QFN/LGA at 0.5 mm or below, or BGA).

2.4. Segregating Functional Regions

The board layout separates functional domains to minimize coupling between noise sources and sensitive circuits:

1. Keep the BNC-to-EZO analog links and the EZO-to-MCU serial lines away from motor control and digital switching sections.
2. Place power conversion (buck converter) and motor control (TMC2209 drivers) near the power entry point, so high-current loops stay short.
3. Filter and transient-protect all power and I/O connectors at the board boundary.
4. Group all power and I/O connectors along one edge of the board where possible, to contain cable radiation (Rule 4c).

2.5. Enclosure and Mechanical

The board targets a ~100 mm × 80 mm footprint, which fits standard off-the-shelf enclosures. Recommended specifications:

- **Enclosure:** IP65-rated ABS, approximately 150 × 100 × 70 mm. The IP65 rating keeps moisture and insects out of the electronics.
- **Cable glands:** Use glands for every wire entering the enclosure — probe cables, pump leads, and the PSU input.
- **BNC connectors:** Mount three panel-mount BNC connectors on the enclosure face for the pH, EC, and RTD probes. Panel-mount rather than PCB-mount prevents mechanical stress on the isolation islands if a probe cable is tugged.
- **Optional:** A clear lid panel allows status LED visibility without opening the enclosure.

2.6. Trace Widths

The trace widths can be calculated using the IPC-2221 empirical formula for external conductors.[5]

$$
\begin{aligned}
I &= k \times \Delta T^{0.44} \times A^{0.725} \\
\text{where } \mathrm{I} &= \text{current } [\mathrm{A}] \\
k &= 0.048 \text{ for outer layer, } \text{ or } 0.024 \text{ for inner layer} \\
\Delta T &= \text{allowable temperature increase } [^\circ \mathrm{C}] \\
A &= \text{cross sectional area } [\mathrm{mil}^2] = \text{width}_{\text{mil}} \times \text{thickness}_{\text{mil}} \\
\text{thickness}_{\text{mil}} &= 1.37\text{mil for 1oz Cu, } \text{ or } 2.74\text{mil for 2oz Cu}
\end{aligned} \tag{1}
$$

The table below uses a conservative $\Delta T = 10^\circ\mathrm{C}$ (IPC-2221 permits 20°C for most PCB classes). The inner-layer widths assume 1 oz copper; the outer-layer widths assume 2 oz copper per §2.3. Power nets are routed on the outer layers in this stack-up (§2.2), so the inner-layer column is reference-only — a 200 mil inner trace at 1 oz carries only ~3.9 A, which is **not** sufficient for the 6.5 A peak on the 24 V input. If any power net must be routed internally, size from the external column or widen accordingly.

Net	Target Current	Internal Trace Width	External Trace Width	Rationale
24V input (PSU→TVS→RPP)	6.5A (Peak)	5.0mm (200mil)	2.0mm (80mil)	Reduce sag
24V main pump	1.2A (Continuous)	1.0mm (40mil)	0.4mm (15mil)	Manage heat
24V each dosing pump	1.53A (Peak)	1.0mm (40mil)	0.4mm (15mil)	Lower inductance
24V ATO valve	0.3A (Peak)	0.2mm (8mil)	0.2mm (8mil)	Fab minimum
5V rail (post-buck)	0.75A (Peak)	0.5mm (20mil)	0.2mm (8mil)	Stable power
3.3V rail (post-LDO)	0.15A (Peak)	0.2mm (8mil)	0.2mm (8mil)	Fab minimum

2.7. PCB Layout Strategy

- **Star power distribution** — Run a dedicated pair of 24V traces from the power entry connector directly to the stepper section, and a separate pair to the logic regulator. Do not daisy-chain power from the motors to the sensors.
- **Via stitching for high-current transitions** — When the 24V rail transitions between layers, use at least 3–4 vias per 2A connection. A single standard 10 mil via carries only 0.5–1A before excessive heating.
- **Antenna keep-out** — The return plane must not extend under the ESP32-C6 antenna keep-out area to ensure proper wireless performance.

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[4] Ralph Morrison, Fast Circuit Boards – Energy Management, Wiley, 2018.
[5] Eric Bogatin, Signal Integrity – Simplified, 3rd edition, Prentice-Hall, 2018.

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1. Signal and Power Integrity, simplified 2nd (2010) - Eric Bogatin ↩
2. PCB Design for Low EMI - Kenneth Wyatt ↩
3. In reality the two fields coexist at every point, but the "leapfrog" picture captures how each sustains the other.

   ↩
4. Derived from the magnetic dipole radiation formula. The full expression includes constants (μ_0, c), but the proportionality to f^2, A, and I captures the design levers. ↩
5. IPC-2221 Trace Width Calculator, Altium PCB Design Guide. ↩